Channel Estimation for Extremely Large-Scale MIMO: Far-Field or Near-Field?

Course: Principles of Wireless Communications

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1 Motivation

Extremely large-scale multiple-input multiple-output (XL-MIMO) uses hundreds of antennas to provide the array gain and spatial resolution needed for high-rate 6G links. Accurate channel state information is essential for precoding, yet the number of radio-frequency chains is much smaller than the number of antennas. Directly estimating every antenna coefficient would therefore require excessive pilot overhead.

Compressed sensing reduces this overhead by assuming that only a few paths are active in an angular Fourier dictionary. Cui and Dai show that this assumption breaks down when the XL-MIMO aperture places users or scatterers in the near field.

2 Far-Field and Near-Field Propagation

The boundary is approximated by the Rayleigh distance

$$Z = \frac{2D^2}{\lambda}, \tag{1}$$

where D is the array aperture and λ is the wavelength. Since Z grows quadratically with D, an XL-MIMO array can have a near-field region extending over a large part of a cell.

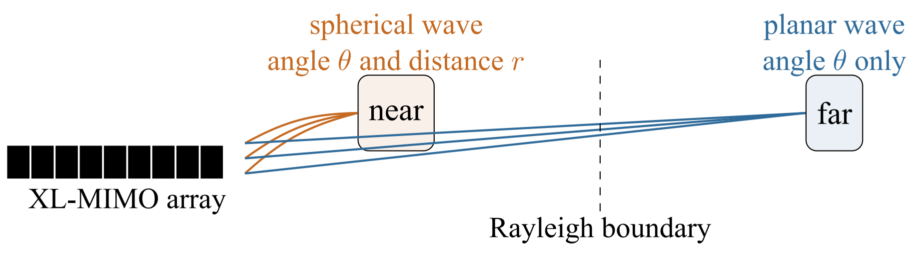

Figure 1: Far-field (planar-wave) and near-field (spherical-wave) propagation separated by the Rayleigh boundary.

Far-field phase changes linearly across the array, so one Fourier vector $\boldsymbol{a}(\theta)$ is sufficient. Near-field phase also depends on distance and requires $\boldsymbol{b}(\theta, r)$. An angle-only dictionary therefore spreads one physical near-field path over many bins.

3 Near-Field Channel Model

For a uniform linear array with antenna index $n = 0, 1, \ldots, N-1$, spacing d, and wavenumber $k_c = 2\pi/\lambda$, the near-field steering vector is

$$\boldsymbol{b}(\theta, r) = \frac{1}{\sqrt{N}} \left[e^{-jk_c(r^{(0)}-r)}, \ldots, e^{-jk_c(r^{(N-1)}-r)} \right]^T, \tag{2}$$

where

$$r^{(n)} = \sqrt{r^2 + \delta_n^2 d^2 - 2r\delta_n d\theta}, \qquad \delta_n = \frac{2n - N + 1}{2}, \tag{3}$$

is the distance between the source and the n-th antenna located at $\delta_n d$, and $\theta \in [-1, 1]$ is the spatial angle. Unlike a Fourier vector, Equation (2) has a nonlinear phase progression that changes with both θ and r.

Figure 2 illustrates the system architecture. The N-element antenna array receives signals from scatterers in the near field. An analog combining matrix $\boldsymbol{A} \in \mathbb{C}^{N_{\mathrm{RF}} \times N}$ reduces the N antenna outputs to N_{RF} RF chains, producing the observation $\overline{\boldsymbol{Y}}$. The polar-domain transform matrix $\boldsymbol{W}$ maps the sparse representation $\boldsymbol{H}^P$ back to the antenna domain. Because the number of RF chains is much smaller than N, compressed sensing is needed to recover the channel from few pilots.

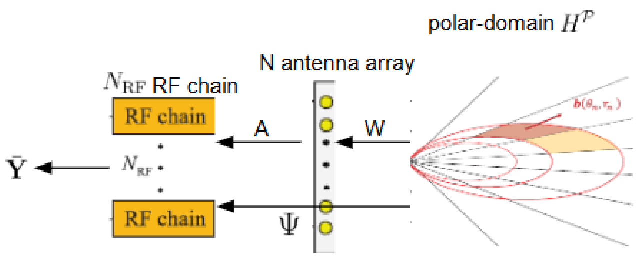

Figure 2: System model: the N-element array, analog combiner $\boldsymbol{A}$, RF chains, and polar-domain transform $\boldsymbol{W}$.

With a limited number of paths, the channel remains compressible even though it is not sparse in angle. The proposed representation is

$$\boldsymbol{h}_m = \boldsymbol{W}\boldsymbol{h}_m^P, \tag{4}$$

where every column of $\boldsymbol{W}$ is a sampled steering vector $\boldsymbol{b}(\theta_p, r_p)$ and $\boldsymbol{h}_m^P$ is sparse in the polar domain. The subscript m denotes a subcarrier; the path locations are shared across subcarriers, which enables simultaneous recovery.

4 From Energy Spread to Polar Sparsity

Figure 3 captures the key motivation. In the angular domain, a far-field path forms a narrow peak, while a near-field path occupies many bins. In the polar domain, each path is represented by an angle–distance pair and the channel again contains only a few significant coefficients. This restored sparsity is what makes low-pilot compressed sensing practical.

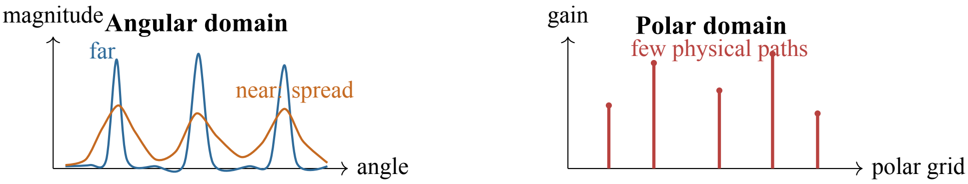

Figure 3: Energy spread and polar sparsity.

5 Design of the Polar-Domain Matrix W

The exact spherical distance makes direct grid design difficult. The Fresnel approximation separates its phase into

$$r^{(n)} - r \approx -\delta_n d\theta + \frac{\delta_n^2 d^2(1-\theta^2)}{2r}. \tag{5}$$

The linear term describes angle; the quadratic term contains the coupled quantity $(1-\theta^2)/r$. This yields the two sampling rules below.

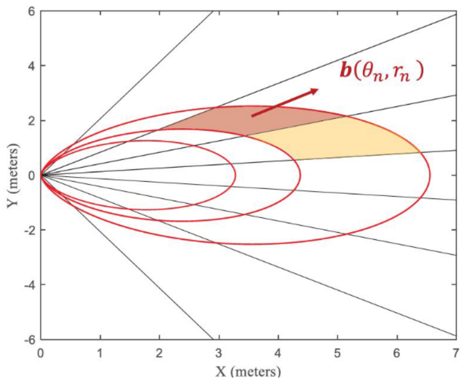

Figure 4: The polar-domain grid produced by the design: distance rings with uniformly sampled angles and steering vectors $\boldsymbol{b}(\theta_n, r_n)$.

Figure 4 shows the resulting polar-domain grid in physical coordinates. Each elliptical contour corresponds to a distance ring on which angles are sampled uniformly. The steering vector $\boldsymbol{b}(\theta_n, r_n)$ points to a specific angle–distance pair. Closer rings are spaced more densely because the sampling is uniform in inverse distance $1/r$, which ensures low mutual coherence between adjacent columns of $\boldsymbol{W}$.

5.1 Angular Sampling on Distance Rings

Locations satisfying

$$\frac{1-\theta^2}{r} = \frac{1}{\rho} \tag{6}$$

have the same quadratic phase and lie on a *distance ring* ρ. Within one ring, coherence is governed by the same linear phase as in a far-field Fourier dictionary. Therefore the angles are sampled uniformly:

$$\theta_n = \frac{2n - N + 1}{N}, \qquad n = 0, 1, \ldots, N-1. \tag{7}$$

5.2 Nonuniform Distance Sampling

For vectors at the same angle, coherence depends on the difference between inverse distances. To keep adjacent columns below a chosen coherence threshold, the sampled distance at angle θ follows

$$r_s(\theta) = \frac{1}{s} Z_\Delta (1-\theta^2), \qquad s = 0, 1, 2, \ldots, \tag{8}$$

where Z_Δ is determined by the array, wavelength, and desired coherence threshold. The case $s = 0$ denotes an infinite-distance far-field ring. Thus distance is sampled uniformly in inverse distance, not uniformly in metres.

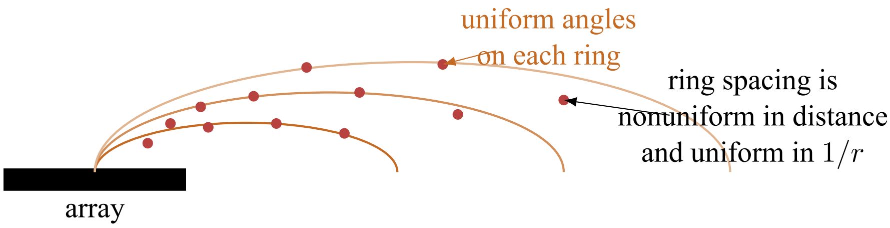

Figure 5: Conceptual distance-ring sampling used to construct $\boldsymbol{W}$.

5.3 Matrix Construction

Each ring produces

$$\boldsymbol{W}_s = [\boldsymbol{b}(\theta_0, r_{s,0}), \ldots, \boldsymbol{b}(\theta_{N-1}, r_{s,N-1})], \tag{9}$$

and $\boldsymbol{W} = [\boldsymbol{W}_0, \ldots, \boldsymbol{W}_{S-1}] \in \mathbb{C}^{N \times NS}$. The $s = 0$ far-field ring makes the Fourier dictionary a special case.

6 Near-Field Channel Estimation Algorithms

After analog combining, the noise is colored. The paper first applies a Cholesky-based pre-whitening transform and obtains the multi-subcarrier model

$$\overline{\boldsymbol{Y}} = \overline{\boldsymbol{\Phi}} \boldsymbol{H}^P + \overline{\boldsymbol{N}}, \qquad \overline{\boldsymbol{\Phi}} = \boldsymbol{D}^{-1} \boldsymbol{A} \boldsymbol{W}. \tag{10}$$

The rows of $\boldsymbol{H}^P$ share a common sparse support because path angles and distances are common across subcarriers.

6.1 On-Grid P-SOMP

P-SOMP extends simultaneous OMP from the angular dictionary to $\boldsymbol{W}$. It proceeds as follows:

1. construct $\boldsymbol{W}$ and pre-whiten the measurements;
2. initialize the residual and an empty support set;
3. select the polar atom with the largest summed correlation across subcarriers;
4. estimate gains by least squares and update the residual;
5. repeat for the detected paths and synthesize the channel.

Because $s = 0$ represents infinite distance, P-SOMP also works for far-field paths. Its weakness is grid mismatch: real angles and distances are continuous and rarely coincide exactly with sampled atoms.

6.2 Off-Grid P-SIGW

P-SIGW performs off-grid near-field channel estimation. Its optimization target is

$$\min_{\hat{\boldsymbol{\theta}},\hat{\boldsymbol{r}}} \mathcal{L}(\hat{\boldsymbol{\theta}},\hat{\boldsymbol{r}}) = -\mathrm{Tr}\left\{\overline{\boldsymbol{Y}}^H \boldsymbol{P}(\hat{\boldsymbol{\theta}},\hat{\boldsymbol{r}})\overline{\boldsymbol{Y}}\right\}, \tag{11}$$

and the algorithm proceeds in three phases.

Phase 1: Initialization. Obtain the initial distances $\hat{\boldsymbol{r}}^0 = [\hat{r}_1^0, \hat{r}_2^0, \ldots, \hat{r}_{\hat{L}}^0]$ and angles $\hat{\boldsymbol{\theta}}^0 = [\hat{\theta}_1^0, \hat{\theta}_2^0, \ldots, \hat{\theta}_{\hat{L}}^0]$ from P-SOMP, then compute the polar-domain gain $\hat{\boldsymbol{G}} = \hat{\boldsymbol{H}}_r^{\mathcal{P}}$ and grid $\hat{\boldsymbol{r}}^0$, $\hat{\boldsymbol{\theta}}^0$.

Phase 2: Refinement. For $n = 1, 2, \ldots, N_{\text{iter}}$, update angle and distance separately using Armijo backtracking line search:

$$\hat{\theta}^n = \hat{\theta}^{n-1} - l_1 \nabla_{\hat{\theta}} \mathcal{L}(\hat{\theta}, \hat{r}^{n-1})\big|_{\hat{\theta}=\hat{\theta}^{n-1}}, \tag{12}$$

$$\frac{1}{\hat{r}^n} = \frac{1}{\hat{r}^{n-1}} - l_2 \nabla_{\frac{1}{\hat{r}}} \mathcal{L}(\hat{\theta}^n, \hat{r})\big|_{\hat{r}=\hat{r}^{n-1}}, \tag{13}$$

where l_1 and l_2 are the step lengths chosen by the Armijo rule. After each iteration, update the path gains via

$$\hat{\boldsymbol{G}}^{\text{opt}} = \widetilde{\boldsymbol{\Psi}}^{\dagger}(\hat{\boldsymbol{\theta}},\hat{\boldsymbol{r}})\overline{\boldsymbol{Y}}. \tag{14}$$

Phase 3: Channel synthesis. Construct the estimated channel as $\hat{\boldsymbol{H}} = [\boldsymbol{b}(\hat{\theta}_1^n, \hat{r}_1^n), \boldsymbol{b}(\hat{\theta}_2^n, \hat{r}_2^n), \ldots, \boldsymbol{b}(\hat{\theta}_{\hat{L}}^n, \hat{r}_{\hat{L}}^n)]$ and return $\hat{\boldsymbol{H}}$.

7 Simulation Results

The distance experiment uses $N = 256$ antennas, a 100 GHz carrier, SNR $= 10$ dB, and pilot length $P = 32$. The corresponding Rayleigh distance is about 100 m. Figure 6 redraws the principal trends from Fig. 10 of the paper rather than reproducing its image.

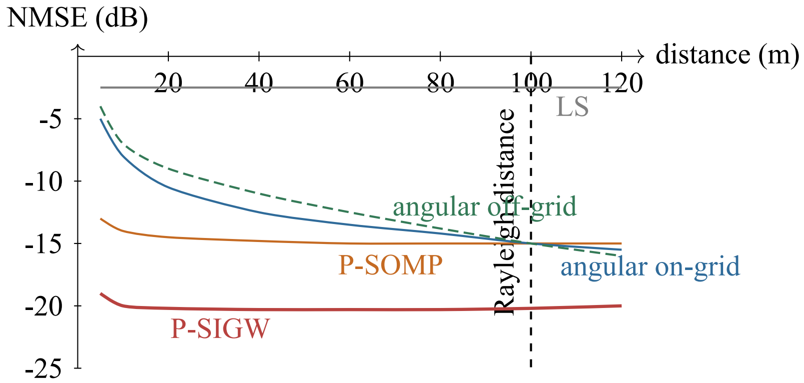

Figure 6: Trend-focused redraw of NMSE versus distance.

8 Interpretation

Inside the Rayleigh distance, angular-domain methods suffer severe model mismatch because a spherical path is not sparse in angle. Their NMSE improves only as distance increases and the wavefront becomes nearly planar. P-SOMP is much less sensitive to distance because its dictionary contains the required angle–distance atoms. P-SIGW is best because it also removes the polar-grid quantization error.

The other experiments in the paper support the same conclusion. As SNR increases, near-field polar methods continue improving while mismatched angular methods approach an error floor. In the far-field setting, the polar methods match the angular baselines and become slightly better at high SNR, because the Fresnel approximation is more accurate than the planar-wave one. Increasing pilot length improves all compressed-sensing estimators, but the gain is much larger when the representation matches the propagation physics.

Q&A

Question. While these two algorithms enhance performance, do they also introduce higher computational complexity compared to existing works? Furthermore, where does the primary complexity bound lie?

Answer.

1. Do they introduce higher computational complexity?

- **P-SOMP:** Overall complexity $\mathcal{O}(\hat{L}PN_{\text{RF}}NSM)$. Compared to the far-field on-grid algorithm SWOMP, its complexity is S times that of SWOMP, where S is the number of sampled distance rings.
- **P-SIGW:** Overall complexity $\mathcal{O}\big(\hat{L}PN_{\text{RF}}NSM + N_{\text{iter}}(P^2N_{\text{RF}}^2M + PN_{\text{RF}}M^2)\big)$. Compared to the far-field off-grid algorithm SS-SIGW-OLS, **the refinement-stage complexity is exactly the same**. The extra cost comes solely from the initialization stage, which runs P-SOMP and thus carries the same S-fold overhead.

Summary: Although higher complexity is introduced, the non-uniform distance-sampling method is highly efficient—S is generally very small (4 to 6 in the simulations)—so this overhead is acceptable.

2. Where does the primary complexity bound lie?

- **P-SOMP bottleneck.** The dominant operation is computing the correlation matrix $\mathbf{\Gamma} = \bar{\mathbf{\Psi}}^H\mathbf{R}$ at each of the $\hat{L}$ iterations. Because XL-MIMO makes N very large and the polar-domain grid covers both angle and distance, the transform matrix $\bar{\mathbf{\Psi}} \in \mathbb{C}^{PN_{\text{RF}} \times NS}$ has NS columns. The bottleneck is therefore $\mathcal{O}(\hat{L}PN_{\text{RF}}NSM)$, the product of paths $\hat{L}$, pilot length P, RF chains N_{RF}, polar-domain grid size NS, and subcarriers M.
- **P-SIGW bottleneck.** The refinement stage dominates because $\hat{L}$ is small. Each iteration computes the gradients of the objective with respect to the angles $\hat{\boldsymbol{\theta}}$ and distances $\hat{\mathbf{r}}$, whose cost is $\mathcal{O}(P^2N_{\text{RF}}^2M + PN_{\text{RF}}M^2)$ per iteration. After N_{iter} iterations the refinement cost is $\mathcal{O}\big(N_{\text{iter}}(P^2N_{\text{RF}}^2M + PN_{\text{RF}}M^2)\big)$. The two terms grow quadratically with the effective observation dimension PN_{RF} and with the subcarrier count M, respectively.